

Notes: Olley and Pakes (Econometrica, 1996)

The Dynamics of Productivity in the Telecommunications Equipment Industry

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1 Big picture

- **Question.** How can we estimate production functions when firms choose inputs knowing their own productivity and when low-productivity firms are more likely to exit the sample?
- **Setting.** The paper studies the U.S. telecommunications equipment industry during 1974–1987, a period of major technological change and deregulation surrounding the Carterphone decisions, the FCC certification program, and the 1982 AT&T consent decree.
- **Core challenge.** Productivity is observed by the firm but not by the econometrician. It affects both current input choices and the decision to shut down. This creates *simultaneity bias* in input coefficients and *selection bias* from exit.
- **Main theoretical result.** The paper develops a semiparametric estimator that uses a monotone investment rule to proxy for the firm’s productivity and then uses survival probabilities to correct for selection among surviving plants.
- **Main empirical result.** Traditional OLS and within estimators are badly biased. In the preferred three-step estimator, the labor coefficient is lower and the capital coefficient is much higher than in balanced-panel or standard full-sample regressions.
- **Main productivity result.** Post-deregulation productivity growth does not come from higher average plant productivity or better within-firm variable-input allocation. It comes mainly from a reallocation of capital and output toward more productive plants.
- **Why this paper matters.** This is the canonical paper that made entry, exit, and endogenous input choice central to production-function estimation, and it also introduced the famous Olley-Pakes decomposition of aggregate productivity into an average term and a covariance term.

2 Data and measurement (key objects)

2.1 Observed plant-level panel

- **Data source.** The data come from the Longitudinal Research Database maintained by the U.S. Census Bureau. The panel combines Census of Manufactures years and Annual Survey of Manufactures years.
- **Industry coverage.** The sample is built mainly from SIC 3661 (telephone and telegraph apparatus) plus a subset of plants from SIC 3663 that produce communications equipment relevant to telecommunications.
- **Main period for estimation.** The estimation sample runs from 1974 to 1987, though the paper uses earlier census years for background facts on entry, exit, and industry structure.
- **Unit of observation.** A plant-year.
- **Observed variables.** Real value added, labor input, capital input, plant age, investment, shipments, employment, and indicators of survival.
- **Real output.** Value added is real shipments net of materials, using four-digit industry deflators.

- **Labor input.** Labor is constructed as total labor compensation divided by the plant’s production-worker wage rate, so it is effectively an hours-style measure.
- **Capital input.** Capital is built with a perpetual-inventory method, allowing separately for structures and equipment, used capital purchases, and rented capital.

2.2 Unobservables and timing

- **Productivity.** The key unobservable is plant productivity, denoted ω_{it} , which is known to the firm when it chooses inputs and whether to continue in operation.
- **Ex post shock.** The production function also contains a shock η_{it} that is not forecastable during the period in which labor can be adjusted.
- **Timing.** Capital is a dynamic state variable. At the beginning of each period, the plant observes its current productivity, decides whether to exit, chooses labor, and chooses investment for the next period.
- **Markov structure.** Productivity follows a first-order Markov process, so current productivity is informative about future productivity and therefore about survival.
- **Economic implication.** Since low-productivity plants are more likely to exit, any estimator that conditions on survival without modeling it will distort the production-function coefficients.

2.3 Empirical applications

The paper uses the estimated production function for four related tasks:

- compare balanced-panel, full-sample, within, and semiparametric estimators of production-function coefficients;
- measure aggregate industry productivity growth over time;
- study whether variable inputs are efficiently allocated conditional on the existing distribution of fixed factors;
- decompose aggregate productivity into average productivity and the covariance between plant productivity and market share, and relate this to entry, exit, and capital reallocation.

3 Models and empirical specifications (all equations + interpretation)

3.1 Dynamic firm behavior and state variables

The plant’s state variables are age a_t , capital k_t , and productivity ω_t . Capital and age evolve as

$$k_{t+1} = (1 - \delta)k_t + i_t, \quad a_{t+1} = a_t + 1. \quad (1)$$

Productivity evolves according to an exogenous Markov process,

$$F_\omega = \{F(\omega' \mid \omega), \omega \in \Omega\}. \quad (2)$$

The incumbent firm's Bellman equation can be written as

$$V_t(\omega_t, a_t, k_t) = \max \left\{ \Phi, \sup_{i_t \geq 0} \pi_t(\omega_t, a_t, k_t) - c(i_t) + \beta \mathbb{E}[V_{t+1}(\omega_{t+1}, a_{t+1}, k_{t+1}) \mid J_t] \right\}, \quad (3)$$

where Φ is the sell-off value if the plant exits.

This problem implies an exit rule and an investment demand equation,

$$\chi_t = \begin{cases} 1 & \text{if } \omega_t \geq \underline{\omega}_t(a_t, k_t), \\ 0 & \text{otherwise,} \end{cases} \quad (4)$$

$$i_t = i_t(\omega_t, a_t, k_t). \quad (5)$$

- **Interpretation.** Firms with higher productivity are more likely to stay alive and to invest more.
- **Why this matters econometrically.** Exit is endogenous, and investment is informative about the unobserved productivity state.

3.2 Production function and the two sources of bias

The paper assumes a Cobb-Douglas production function in logs:

$$y_{it} = \beta_0 + \beta_a a_{it} + \beta_k k_{it} + \beta_l l_{it} + \omega_{it} + \eta_{it}. \quad (6)$$

- **Simultaneity.** Labor is a variable input that can respond to ω_{it} , so l_{it} is positively correlated with the unobserved productivity term. This biases the labor coefficient upward in naive OLS.
- **Selection.** Plants survive only when ω_{it} is high enough relative to age and capital. Because higher-capital plants can survive at lower productivity levels, conditioning on survival tends to generate a negative bias in the capital coefficient.
- **Balanced-panel problem.** Restricting the sample to plants that survive the entire period intensifies the selection bias because the sample is mechanically chosen on long-run survival.

3.3 Investment inversion and the partially linear first stage

The key identifying restriction is that, for plants with positive investment, investment is strictly increasing in productivity conditional on age and capital. Therefore, on the subset with $i_t > 0$, the investment rule can be inverted as

$$\omega_{it} = h_t(i_{it}, a_{it}, k_{it}). \quad (7)$$

Substituting this into the production function gives the partially linear equation

$$y_{it} = \beta_l l_{it} + \phi_t(i_{it}, a_{it}, k_{it}) + \eta_{it}, \quad (8)$$

where

$$\phi_t(i_{it}, a_{it}, k_{it}) = \beta_0 + \beta_a a_{it} + \beta_k k_{it} + h_t(i_{it}, a_{it}, k_{it}). \quad (9)$$

- **What this stage identifies.** Because η_{it} is orthogonal to current inputs, the partially linear regression identifies the labor coefficient β_l .
- **What it does not identify.** It does *not* separately identify β_a and β_k , because age and capital matter both directly in production and indirectly through the investment function.

3.4 Survival probabilities and selection correction

Let the probability of surviving from period t to $t + 1$ be

$$P_t(i_t, a_t, k_t) = \mathbb{P}\{\chi_{t+1} = 1 \mid i_t, a_t, k_t\}. \quad (10)$$

Because investment is monotone in productivity, the survival probability can also be expressed as a function of the current productivity state.

The crucial conditional expectation is

$$\mathbb{E}[y_{t+1} - \beta_l l_{t+1} \mid a_{t+1}, k_{t+1}, \chi_{t+1} = 1, J_t] = \beta_0 + \beta_a a_{t+1} + \beta_k k_{t+1} + \mathbb{E}[\omega_{t+1} \mid \omega_t, \chi_{t+1} = 1]. \quad (11)$$

The last term can be summarized by a two-index function,

$$\beta_0 + \beta_a a_{t+1} + \beta_k k_{t+1} + g(P_t, \omega_t). \quad (12)$$

Using the first-stage estimate of ϕ_t , this becomes the estimable second-stage equation

$$y_{t+1} - \beta_l l_{t+1} = \beta_a a_{t+1} + \beta_k k_{t+1} + g(P_t, \phi_t - \beta_a a_t - \beta_k k_t) + \xi_{t+1} + \eta_{t+1}, \quad (13)$$

where ξ_{t+1} is the innovation in productivity net of the selection correction.

- **Interpretation.** The selection term depends on both the current productivity state and the survival threshold, and the survival probability acts like a propensity-score control for the latter.
- **Why it works.** Once current productivity is proxied by investment and the survival threshold is summarized by the survival probability, the remaining variation can identify capital and age.

3.5 Estimator implementation

The paper implements a three-step semiparametric estimator.

- **Step 1:** Estimate the partially linear model with a fourth-order polynomial in (i_t, a_t, k_t) , allowing this polynomial to vary across four regulatory periods. This identifies β_l .
- **Step 2:** Estimate the survival probability $P_t(i_t, a_t, k_t)$ using both kernel and series methods. The preferred version is a bias-reducing kernel estimator that allows the selection rule to differ across the same four regulatory periods.
- **Step 3:** Estimate (β_a, β_k) by nonlinear least squares or kernel methods in the second-stage equation above.

The preferred estimates use a series first stage, a kernel survival estimator, and a kernel third stage. They imply

$$\hat{\beta}_l = 0.608, \quad \hat{\beta}_k = 0.355, \quad \hat{\beta}_a \approx 0.01. \quad (14)$$

- **Interpretation.** Relative to standard estimators, labor is much less important and capital much more important once simultaneity and selection are handled.

3.6 Aggregate productivity, static efficiency, and the OP decomposition

Using the preferred production-function estimates, plant-level productivity is

$$p_{it} = \exp(y_{it} - \beta_l l_{it} - \beta_k k_{it} - \beta_a a_{it}). \quad (15)$$

Aggregate industry productivity is the output-share-weighted average of plant productivity,

$$P_t = \sum_{i=1}^{N_t} s_{it} p_{it}, \quad (16)$$

where s_{it} is plant i 's output share.

The paper then decomposes aggregate productivity as

$$P_t = \bar{p}_t + \sum_{i=1}^{N_t} (s_{it} - \bar{s}_t)(p_{it} - \bar{p}_t), \quad (17)$$

where $\bar{p}_t$ is the unweighted mean of plant productivity and $\bar{s}_t$ is the unweighted mean output share.

- **Interpretation.** Aggregate productivity equals average plant productivity plus a covariance term. The covariance term is high when more output is concentrated in more productive plants.
- **This is the Olley-Pakes decomposition.** Later work made this decomposition famous, but it originates here as a way to isolate whether industry-level productivity growth comes from improving average plants or from reallocating market share.

The paper also studies a static variable-cost-efficiency index. Plant-level variable cost is

$$C(Y_i, K_i, a_i, p_i, w_i) = \min_{L_i} w_i L_i \quad \text{s.t.} \quad Y_i \leq L_i^{\beta_l} K_i^{\beta_k} e^{\beta_a a_i} p_i, \quad (18)$$

and the minimum total variable cost of producing industry output is

$$\min_{\{Y_i\}} \sum_i C(Y_i, K_i, a_i, p_i, w_i) \quad \text{s.t.} \quad \sum_i Y_i = Y^{\text{industry}}. \quad (19)$$

- **Interpretation.** This index asks whether labor is efficiently allocated across plants *conditional on the existing distribution of capital, age, and productivity*. It is deliberately different from the OP covariance term, which reflects the broader allocation of output and capital across heterogeneous plants.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in the standard approach

- **Balanced panels are selected samples.** A plant must be productive enough to survive throughout the sample to appear in a balanced panel. This selection is correlated with the unobserved productivity term and especially biases the capital coefficient downward.

- **Full-sample OLS still has simultaneity bias.** Even when entrants and exiters are kept in the data, labor remains positively correlated with current productivity because firms adjust labor knowing their productivity.
- **Within estimators are not enough.** The within estimator helps only under very restrictive assumptions in which the unobserved productivity component is fixed over time. That is not plausible in a restructuring industry with major entry, exit, and technological change.

4.2 What identifies the OP estimator

- **Single unobserved state variable.** The model assumes that one scalar state, productivity, is sufficient to summarize the firm-specific unobservable affecting both investment and exit.
- **Monotone investment rule.** For plants with positive investment, investment must be increasing in productivity conditional on age and capital, so that the investment rule can be inverted.
- **Timing.** Labor can react to current productivity, but capital is predetermined and the ex post shock η_{it} is unforecastable. This timing is what makes the first stage informative about labor and the second stage informative about capital.
- **Selection correction through survival probabilities.** The propensity-score-style control $P_t(i_t, a_t, k_t)$ summarizes the relevant survival threshold and lets the estimator control for the fact that the observed transition from t to $t + 1$ is conditional on not having exited.

4.3 What makes the estimates empirically credible

- **Better-fitting survival equations.** The kernel survival estimator with different selection equations across regulatory regimes fits actual survival better than the simpler series alternatives.
- **Internal specification tests.** When contemporaneous labor is added to the final equation, its coefficient is insignificant. When current age and capital are added directly, they are also insignificant. This supports the internal restrictions imposed by the model.
- **Robustness across series and kernel methods.** The final capital coefficient is almost identical across the series and kernel third-stage estimators.
- **Subsample robustness.** The same qualitative pattern – lower labor and higher capital than OLS – appears in early and late subsamples and in switchmaker and nonswitchmaker subsamples.

4.4 Relation to the literature

- **Marschak-Andrews problem.** The paper directly addresses the classic simultaneity problem in production-function estimation.
- **Selection problem.** It is one of the first papers to build a production-function estimator that explicitly models endogenous exit.
- **Dynamic industry models.** The behavioral framework borrows from Ericson-Pakes and Hopenhayn-Rogerson style models of firm dynamics, but the estimating procedure is much lighter and semiparametric.

- **Heckman-style selection correction.** The survival correction resembles a dynamic single-index selection problem, with the survival probability acting like a propensity score.

5 Tables and figures

5.1 Table I: Characteristics of the data

Read:

- The industry expanded dramatically before the early-1980s shakeout. Plants rose from 164 in 1967 to 584 in 1987, and firms rose from 131 to 481.
- Real shipments increased from 8.179 billion 1982 dollars in 1967 to 22.413 billion in 1987.
- Employment grew from 162,402 in 1967 to a peak of 222,058 in 1982, then fell to 184,178 by 1987.
- The table sets up the basic industry fact: strong long-run growth together with intense restructuring.

TABLE I
CHARACTERISTICS OF THE DATA

Year	Plants	Firms	Shipments (billions 1982 \$)	Employment
1963	133	104	5.865	136899
1967	164	131	8.179	162402
1972	302	240	11.173	192248
1977	405	333	13.468	192259
1982	473	375	20.319	222058
1987	584	481	22.413	184178

TABLE II
BELL COMPANY EQUIPMENT PROCUREMENT
(PERCENT PURCHASED FROM WESTERN ELECTRIC)

1982	1983	1984	1985	1986 ^E
92.0	80.0	71.8	64.2	57.6

^E Estimated for 1986.

Source: NTIA (1988, p. 336, and discussion pp. 335–337).

5.2 Tables II–IV: Deregulation, entry, and exit

Read:

- Table II shows how the 1982 consent decree broke Western Electric’s dominance. The Bell companies’ equipment purchases from Western Electric fell from 92.0% in 1982 to 57.6% by 1986.
- Table III shows that entry was massive: by 1987, 79% of plants and 87% of firms active in the industry had entered since 1972. Those post-1972 entrants accounted for 32.8% of shipments and 36.0% of employment in 1987.
- Table IV shows that exit was equally important. Of the plants active in 1972, 60% did not survive to 1987; of the plants active in 1982, 41.2% did not survive to 1987.
- These tables explain why a balanced panel is such a bad idea here: it throws away the central margins of adjustment.

TABLE III
ENTRANTS ACTIVE IN 1987

	Number	Share of Number Active in 1987 (%)	Share of 1987 Shipments (%)	Share of 1987 Employment (%)
Plants: New since 1972	463	79.0	32.8	36.0
Firms: New since 1972	419	87.0	30.0	41.4
Plants: New since 1982	306	52.0	12.0	13.5
Firms: New since 1982	299	60.1	19.4	27.5

TABLE IV
INCUMBENTS EXITING BY 1987

	Number	Share of Number Active in Base Year (%)	Share of Shipments in Base Year (%)	Share of Employment in Base Year (%)
Plants active in 1972 but not in 1987	181	60.0	40.2	39.0
Firms active in 1972 but not in 1987	169	70.0	13.8	12.1
Plants active in 1982 but not in 1987	195	41.2	26.0	24.1
Firms active in 1982 but not in 1987	184	49.1	17.3	16.1

5.3 Table V: Correlation coefficients between predicted survival probabilities

Read:

- The paper compares several estimators of the survival probability.
- The kernel estimator that allows different selection equations across the four regulatory regimes is the best-fitting one and becomes the preferred choice in the final estimator.
- The broader lesson is that the survival rule itself changes with regulation, so a single pooled exit equation is too restrictive.

TABLE V
CORRELATION COEFFICIENTS BETWEEN VARIOUS PREDICTED SURVIVAL PROBABILITIES AND χ_{t+1}

	χ_{t+1}	PHAT1	PHAT2	PHAT3	PHAT4
χ_{t+1}	1.00	.285	.350	.102	.218
PHAT1	.285	1.00	.671	.398	.324
PHAT2	.350	.671	1.00	.215	.583
PHAT3	.102	.398	.215	1.00	.483
PHAT4	.218	.324	.583	.483	1.00

Notes: (1) χ_{t+1} is a 0,1 random variable that takes the value 0 when a plant closes.
(2) PHAT1 and PHAT2 are the kernel estimates. PHAT1 is estimated over the entire data set, and PHAT2 is estimated separately for the four time periods: 1974–1977, 1978–1980, 1981–1983, and 1984–1987.
(3) PHAT3 and PHAT4 are the probit estimates. PHAT3 has no time dummies, and PHAT4 is estimated with time period dummies corresponding to the periods in note (2), and these dummies are interacted with i_j , k_j , and d_j .

TABLE VI
ALTERNATIVE ESTIMATES OF PRODUCTION FUNCTION PARAMETERS^a
(STANDARD ERRORS IN PARENTHESES)

Sample:	Balanced Panel		Full Sample ^{c,d}						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	Nonparametric F_w	
Estimation Procedure	Total	Within	Total	Within	OLS	Only P	Only h	Series	Kernel
Labor	.851 (.039)	.728 (.049)	.693 (.019)	.629 (.026)	.628 (.020)				.608 (.027)
Capital	.173 (.034)	.067 (.049)	.304 (.018)	.150 (.026)	.219 (.018)	.355 (.02)	.339 (.03)	.342 (.035)	.355 (.058)
Age	.002 (.003)	-.006 (.016)	-.0046 (.0026)	-.008 (.017)	-.001 (.002)	-.003 (.002)	.000 (.004)	-.001 (.004)	.010 (.013)
Time	.024 (.006)	.042 (.017)	.016 (.004)	.026 (.017)	.012 (.004)	.034 (.005)	.011 (.01)	.044 (.019)	.020 (.046)
Investment	—	—	—	—	.13 (.01)	—	—	—	—
Other Variables	—	—	—	—	—	Powers of P	Powers of h	Full Polynomial in P and h	Kernel in P and h
# Obs. ^b	896	896	2592	2592	2592	1758	1758	1758	1758

^aThe dependent variable in columns (1) to (5) is the log of value added, while in columns (6) to (10), the dependent variable is the log of value added $-b_j \cdot \log(\text{labor})$.

^bThe number of observations in the balanced panels of regressions 1 and 2 are the observations for those plants that have continuous data over the period, with zero investment observations removed. The 2592 observations used in columns (3), (4), and (5) are all observations in the full sample except those with zero investment. Approximately 8% of the full data set had observations with zero investment. Columns (6) to (10) have fewer observations because the sampling procedures for the Annual Survey of Manufactures forced us to drop observations in years 1978, 1983, and the last year, 1987. See note c.

^cThe number of observations in the last four columns decreases to 1758 because we needed lagged values of some of the independent variables in estimation. This rules out using the first observation on each plant and the first year of the rotating five-year panels that make up the Annual Survey of Manufactures. To check that the difference between the estimates in columns (6)–(9) and those in columns (3)–(5) are not due to the sample, we ran the estimating equations in columns (3)–(5) on the 1758 plant sample and got almost identical results.

^dConsult the text for details of the estimation algorithm for columns (6) to (10).

5.4 Table VI: Alternative estimates of production function parameters

Read:

- In the balanced panel, the OLS estimates are labor 0.851 and capital 0.173; the within estimates are labor 0.728 and capital 0.067. These are the familiar “high labor, tiny capital” results that the paper argues are misleading.
- Simply moving from the balanced panel to the full sample more than doubles the capital coefficient and lowers the labor coefficient by about 20%: full-sample OLS gives labor 0.693 and capital

0.304; full-sample within gives labor 0.629 and capital 0.150.

- The preferred three-step kernel estimator gives labor 0.608 and capital 0.355. Relative to full-sample OLS, labor is about 15% lower; relative to the full-sample within estimator, capital is more than twice as large.
- This table is the paper's main empirical point: once simultaneity and selection are treated structurally, capital matters much more and labor much less than standard estimators suggest.

5.5 Table VII: Specification tests

Read:

- When contemporaneous labor is added to the final estimating equation, its coefficient is statistically insignificant and the capital coefficient stays essentially unchanged at about 0.36.
- When current age and capital are added directly, both coefficients are also insignificant and the core estimates barely move.
- These tests do not prove the model is exactly right, but they show that the paper's main restrictions are not visibly contradicted by the data.

TABLE VII
SPECIFICATION TESTS^a
(STANDARD ERRORS IN PARENTHESES)

	Test 1	Test 2
b_l^b	.608	.608
b_a	.01 (.01)	.01 (.01)
b_k	.36 (.06)	.36 (.06)
b_t	.02 (.05)	.02 (.05)
γ_l	-.01 (.03)	—
γ_a	—	.00 (.02)
γ_k	—	-.01 (.04)

^aSee text for details.

^bThe labor coefficient is taken from the first equation, and is the same in both tests.

TABLE VIII
PRODUCTION FUNCTION PARAMETER ESTIMATES
(STANDARD ERRORS IN PARENTHESES)

	Labor	Capital	Age	Time	# Obs.
1974–1978					
OLS	.78 (.03)	.27 (.03)	-.00 (.003)	.03 (.01)	832
3-Step Procedure ^a	.71 (.05)	.29 (.05)	.01 (.03)	.10 (.19)	578 ^b
1982–1987					
OLS	.62 (.03)	.33 (.03)	-.01 (.002)	-.02 (.010)	1212
3-Step Procedure	.55 (.05)	.40 (.13)	.01 (.02)	-.01 (.11)	729
Switch Makers ^c					
OLS	.79 (.05)	.25 (.05)	-.01 (.003)	.03 (.01)	562
3-Step Procedure	.66 (.07)	.31 (.16)	.01 (.04)	.01 (.16)	387
Non-Switch Makers					
OLS	.67 (.02)	.32 (.02)	-.00 (.002)	.01 (.004)	2030
3-Step Procedure	.59 (.03)	.37 (.05)	.01 (.03)	.04 (.03)	1433

^aThe 3-step estimation procedure is described in detail in the text. The labor coefficient is obtained in the first step using a fourth order polynomial series estimator, the second stage uses a bias reducing kernel to estimate the survival probability, and third stage uses a bias reducing kernel estimator.

^bThe number of observations in the 3-step procedure is the number of observations in the third step of the estimation procedure in which the age and capital coefficient estimates are obtained.

^cSwitch makers are plants that primarily produce switching equipment, and nonswitch makers include all other plants.

5.6 Table VIII: Subsample estimates by period and product type

Read:

- The direction of the correction survives in every subsample. In both 1974–1978 and 1982–1987, the three-step estimator lowers labor and raises capital relative to OLS.
- The same is true when the sample is split into switchmakers and nonswitchmakers.

- So the main result is not driven by one particular era or product category inside the industry.

5.7 Table IX: Industry productivity growth rates

Read:

- In the full sample, annual productivity growth is -27.9% from 1974 to 1975, then 2.0% in 1975–1977, 14.6% in 1978–1980, -8.7% in 1981–1983, and 4.1% in 1984–1987.
- Excluding the sharp 1974–1975 drop, average annual productivity growth in the full sample is 3.2% over 1975–1987.
- The high-growth periods are exactly the periods following regulatory liberalization. The negative 1981–1983 period matches the disruptive reorganization surrounding divestiture.
- The balanced-panel productivity series is materially different, especially after 1978, because it omits the entry and exit that carried much of the restructuring.

TABLE IX
INDUSTRY PRODUCTIVITY GROWTH RATES^a

Time Period	(1) Full Sample	(2) Balanced Panel
1974–1975	–.279	–.174
1975–1977	.020	–.015
1978–1980	.146	.102
1981–1983	–.087	–.038
1984–1987	.041	.069
1974–1987	.008	.020
1975–1987	.032	.036
1978–1987	.034	.047

^aThe numbers in Table IX are annual averages over the various subperiods.

TABLE X
VARIABLE COST EFFICIENCY^a
(MINIMUM COST OF PRODUCTION DIVIDED BY ACTUAL COST OF PRODUCTION)

Years	Total	Interfirm	Intrafirm
1974–1977	.77	.84	.91
1978–1980	.69	.76	.91
1981–1983	.65	.72	.91
1984–1987	.72	.80	.89

^aSee text for details.

5.8 Table X: Variable cost efficiency

Read:

- The total variable-cost-efficiency index falls from 0.77 in 1974–1977 to 0.69 in 1978–1980 and 0.65 in 1981–1983, then recovers somewhat to 0.72 in 1984–1987.
- Almost all movement comes from the *interfirm* component. The intrafirm component stays roughly constant at about 0.89–0.91.
- This means post-deregulation productivity growth did not come from a more efficient allocation of variable factors conditional on the existing distribution of fixed factors. The action must be in the distribution of capital and output across plants.

5.9 Table XI: Decomposition of productivity

Read:

- The unweighted average of plant productivity shows little sustained upward trend after 1975. It fluctuates roughly between 0.66 and 0.84.
- The covariance term, by contrast, rises sharply after the regulatory changes: it reaches 0.12 in 1978, 0.28 in 1980, 0.35 in 1981, and 0.32 in 1987.
- The correlation between capital and productivity also rises over time, from -0.07 in 1974 to 0.10 in 1987.
- This table delivers the main economic message of the paper: productivity growth came mainly from reallocating output and capital toward more productive plants, not from raising the productivity of the average plant.

TABLE XI
DECOMPOSITION OF PRODUCTIVITY^a
(EQUATION (16))

Year	p_t	$\bar{p}_t$	$\Sigma_t \Delta s_{it} \Delta p_{it}$	$\rho(p_t, k_t)$
1974	1.00	0.90	0.01	-0.07
1975	0.72	0.66	0.06	-0.11
1976	0.77	0.69	0.07	-0.12
1977	0.75	0.72	0.03	-0.09
1978	0.92	0.80	0.12	-0.05
1979	0.95	0.84	0.12	-0.05
1980	1.12	0.84	0.28	-0.02
1981	1.11	0.76	0.35	0.02
1982	1.08	0.77	0.31	-0.01
1983	0.84	0.76	0.08	-0.07
1984	0.90	0.83	0.07	-0.09
1985	0.99	0.72	0.26	0.02
1986	0.92	0.72	0.20	0.03
1987	0.97	0.66	0.32	0.10

^aSee text for details.

TABLE XII
PROBIT MODELS OF EXIT PROBABILITIES^a
(STANDARD ERRORS IN PARENTHESES)

	1	2	3
Intercept	-1.39 (.11)	-0.69 (.25)	-0.63 (.25)
Productivity	-0.16 (.06)	-0.15 (.06)	-0.16 (.06)
Age		0.00 (.01)	-0.00 (.01)
Capital		-0.09 (.03)	-0.10 (.03)
D2			-0.37 (.20)
D3			0.10 (.14)
D4			0.47 (.12)
# Obs.	2098	2098	2098
Log Likelihood	-392.2	-387.1	-372.1

^aThe dummy variables are defined as follows: Base period is 1974–1977; D2 = 1 for years 1978 to 1980, 0 otherwise; D3 = 1 for years 1981 to 1983, 0 otherwise; D4 = 1 for years 1984 to 1987, 0 otherwise.

5.10 Table XII: Probit models of exit probabilities

Read:

- Exit is strongly related to plant fundamentals. The probability of exit falls with productivity and with capital, and the productivity effect is larger in magnitude.
- In the specification with both productivity and capital, the coefficients are about -0.16 on productivity and -0.10 on capital.
- The post-1984 dummy is positive and sizable, indicating that conditional on fundamentals, plants are more likely to exit after divestiture.
- This is exactly the mechanism behind the paper's selection correction and its broader productivity story: unproductive plants shrink and shut down, releasing capital and output share to better plants.

6 Short summary

- **Conclusion.** The paper shows that production-function estimation must explicitly account for endogenous exit and endogenous input choice.
- **Identification insight.** Investment can proxy for productivity only when combined with a structural correction for the survival rule; otherwise balanced-panel and standard full-sample estimates are systematically biased.
- **Empirical lesson.** The post-deregulation productivity gains in telecommunications equipment came mostly from reallocation – especially of capital and output toward more productive plants – rather than from higher average productivity or better variable-input allocation within the existing stock of plants.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that selection from exit and simultaneity from endogenous input choice are quantitatively important in plant-level production-function estimation.
- It provides a practical semiparametric estimator that addresses both problems jointly.
- Empirically, it shows that standard methods understate the capital coefficient and overstate the labor coefficient in a setting with substantial restructuring.

7.2 Economic intuition

The basic intuition is simple. Productive plants want to invest and survive; unproductive plants contract and often shut down. If the econometrician ignores this, the sample of observed plants becomes increasingly selected, and the plants that remain are not representative of the original industry. At the same time, labor responds to productivity shocks, so a naive regression mistakes part of productivity for labor’s causal contribution. Once the paper explicitly models these forces, the production function looks much more sensible. The same logic then explains aggregate productivity growth: industry productivity rose not because every plant got better, but because more output and capital ended up in the hands of better plants.

7.3 Takeaway

This paper is a foundation stone for modern empirical work on firm dynamics and misallocation. Its econometric lesson is that you cannot treat entry, exit, and input choice as nuisances in dynamic industries. Its substantive lesson is that deregulation can raise aggregate productivity by changing who produces, not just how production is organized inside the average surviving plant. And its decomposition of aggregate productivity into an average term and a covariance term became one of the most influential tools in the later productivity literature.